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Learning Objectives

- Apply the Net Change Theorem to velocity, rate, and population problems
- Estimate definite integrals using Left, Right, Midpoint, and Trapezoidal Rules
- Evaluate accumulation functions $F(x) = \int_{\blacksquare}^{\blacksquare} f(t) dt$ and find $F'(x)$ by FTC
- Distinguish displacement from total distance using signed vs absolute area

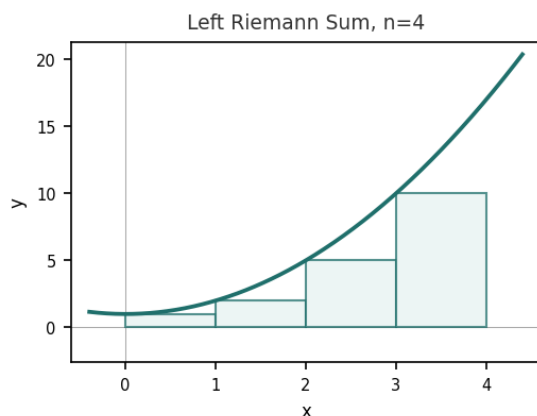
For Riemann sums, show your table of x -values and function values before summing. For total distance, find where $v(t) = 0$ and split the integral.

1. A particle moves with velocity $v(t) = 3t^2 - 12t + 9$ ft/s. Find the net displacement and total distance traveled on $[0, 4]$.

$$\int_0^4 (3t^2 - 12t + 9) dt \quad \text{and} \quad \int_0^4 |v(t)| dt$$

Answer: _____

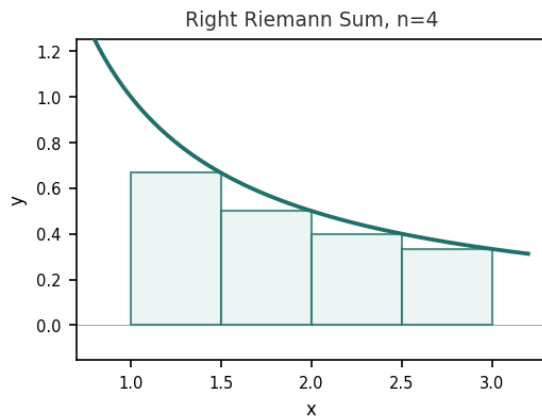
2. Approximate the integral of $f(x) = x^2 + 1$ from $x = 0$ to $x = 4$ using a Left Riemann Sum with $n = 4$ rectangles.



Answer: _____

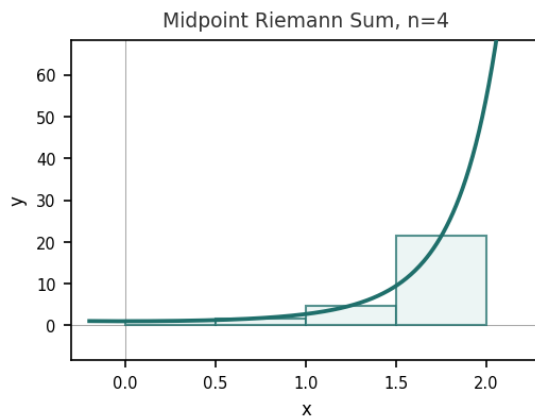


3. Approximate the integral of $f(x) = 1/x$ from $x = 1$ to $x = 3$ using a Right Riemann Sum with $n = 4$ rectangles.



Answer: _____

4. Approximate the integral of $f(x) = e^{(x^2)}$ from $x = 0$ to $x = 2$ using the Midpoint Rule with $n = 4$.



Answer: _____

5. Use the Trapezoidal Rule with $n = 4$ to approximate the integral of $\sin(x)$ from 0 to π .

$$T_4 = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$$

Answer: _____

6. Oil leaks from a pipe at rate $r(t) = 4t - t^2 + 3$ gallons/hour ($0 \leq t \leq 3$). Find the total oil leaked.

$$\int_0^3 (4t - t^2 + 3) dt$$

Answer: _____

7. Let $F(x)$ be defined by the integral below. Find $F(3)$ and $F'(x)$. Interpret each.

$$F(x) = \int_0^x (t^2 - t) dt$$

Answer: _____



8. Evaluate using the Fundamental Theorem of Calculus.

$$\int_1^4 \left(\sqrt{x} - \frac{1}{x} \right) dx$$

Answer: _____

9. A city's population grows at the rate shown below. The population at $t = 0$ is 50,000. Use the Net Change Theorem to find $P(10)$.

$$P'(t) = 400e^{0.02t}, \quad P(10) = P(0) + \int_0^{10} 400e^{0.02t} dt$$

Answer: _____

10. Compute L4, R4, and M4 with $n = 4$ for the integral below. Compare each approximation to the exact value.

$$\int_0^2 x^3 dx, \quad \Delta x = 0.5$$

Answer: _____





Problems 2–4 include Riemann sum diagrams. Problem 1: displacement vs distance — the classic distinction. Problem 7 (accumulation function): connects FTC Part 1 and Part 2.

Solutions

1. A particle moves with velocity $v(t) = 3t^2 - 12t + 9$ ft/s. Find the net displacement and total distance traveled on $[0, 4]$.

$$\int_0^4 (3t^2 - 12t + 9) dt \quad \text{and} \quad \int_0^4 |v(t)| dt$$

→ Antiderivative: $t^3 - 6t^2 + 9t$.

→ Displacement: evaluate $[t^3 - 6t^2 + 9t]$ from 0 to 4 = $64 - 96 + 36 = 4$ ft.

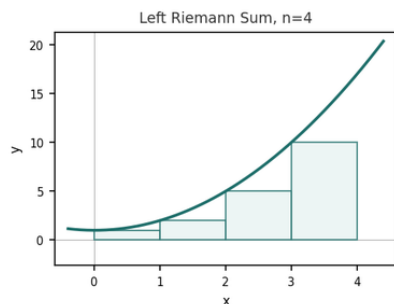
→ $v(t) = 3(t-1)(t-3)$. Zeros at $t=1$ and $t=3$.

→ $v \geq 0$ on $[0, 1]$ and $[3, 4]$; $v \leq 0$ on $[1, 3]$ — must split for total distance.

→ Total distance = $|\text{disp on } [0, 1]| + |\text{disp on } [1, 3]| + |\text{disp on } [3, 4]| = 4 + 4 + 4 \dots$ evaluate each.

Answer: Displacement = 4 ft, Distance = 14 ft

2. Approximate the integral of $f(x) = x^2 + 1$ from $x = 0$ to $x = 4$ using a Left Riemann Sum with $n = 4$ rectangles.



→ $dx = (4-0)/4 = 1$. Left endpoints: $x = 0, 1, 2, 3$.

→ $f(0)=1, f(1)=2, f(2)=5, f(3)=10$.

→ $L_4 = 1 \cdot (1 + 2 + 5 + 10) = 18$.

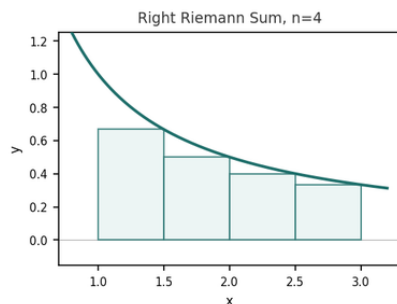
→ Exact value: $[x^3/3 + x]$ from 0 to 4 = $64/3 + 4 = 76/3 \approx 25.33$.

→ L_4 underestimates because f is increasing on $[0, 4]$.

Answer: $L_4 = 1 \cdot [1 + 2 + 5 + 10] = 18$



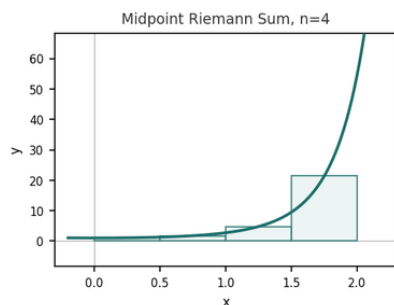
3. Approximate the integral of $f(x) = 1/x$ from $x = 1$ to $x = 3$ using a Right Riemann Sum with $n = 4$ rectangles.



→ $dx = (3-1)/4 = 0.5$. Right endpoints: $x = 1.5, 2.0, 2.5, 3.0$.
 → $f(1.5)=2/3, f(2)=1/2, f(2.5)=2/5, f(3)=1/3$.
 → $R_4 = 0.5*(2/3 + 1/2 + 2/5 + 1/3) = 0.5*(1.900) \approx 0.950$.
 → Exact: $\ln(3) \approx 1.099$. R_4 underestimates (f is decreasing on $[1,3]$).

Answer: $R_4 = \frac{1}{2} \left[\frac{2}{3} + \frac{1}{2} + \frac{2}{5} + \frac{1}{3} \right] \approx 0.95$

4. Approximate the integral of $f(x) = e^{x^2}$ from $x = 0$ to $x = 2$ using the Midpoint Rule with $n = 4$.



→ $dx = 0.5$. Midpoints: $x = 0.25, 0.75, 1.25, 1.75$.
 → $f(0.25) = e^{0.0625} \approx 1.064, f(0.75) = e^{0.5625} \approx 1.755$.
 → $f(1.25) = e^{1.5625} \approx 4.772, f(1.75) = e^{3.0625} \approx 21.38$.
 → $M_4 = 0.5*(1.064 + 1.755 + 4.772 + 21.38) = 0.5 * 28.97 \approx 14.49$.

Answer: $M_4 \approx 0.5[e^{0.0625} + e^{0.5625} + e^{1.5625} + e^{3.0625}] \approx 14.5$

5. Use the Trapezoidal Rule with $n = 4$ to approximate the integral of $\sin(x)$ from 0 to π .

$$T_4 = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$$

→ $dx = \pi/4$. x -values: $0, \pi/4, \pi/2, 3\pi/4, \pi$.
 → $\sin(0)=0, \sin(\pi/4)=\sqrt{2}/2, \sin(\pi/2)=1, \sin(3\pi/4)=\sqrt{2}/2, \sin(\pi)=0$.
 → $T_4 = (\pi/8)[0 + 2(\sqrt{2}/2) + 2(1) + 2(\sqrt{2}/2) + 0] = (\pi/8)(2\sqrt{2}+2) \approx 1.896$.
 → Exact: integral of $\sin(x)$ from 0 to $\pi = 2$. Error ≈ 0.104 .

Answer: $T_4 \approx \frac{\pi}{8} \left[0 + 2 \cdot \frac{\sqrt{2}}{2} + 2 + 2 \cdot \frac{\sqrt{2}}{2} + 0 \right] \approx 1.896$



6. Oil leaks from a pipe at rate $r(t) = 4t - t^2 + 3$ gallons/hour ($0 \leq t \leq 3$). Find the total oil leaked.

$$\int_0^3 (4t - t^2 + 3) dt$$

→ Antiderivative: $2t^2 - t^3/3 + 3t$.

→ At $t=3$: $2(9) - 27/3 + 9 = 18 - 9 + 9 = 18$.

→ At $t=0$: 0.

→ Total oil leaked = 18 gallons.

Answer: $\left[2t^2 - \frac{t^3}{3} + 3t \right]_0^3 = 18$

7. Let $F(x)$ be defined by the integral below. Find $F(3)$ and $F'(x)$. Interpret each.

$$F(x) = \int_0^x (t^2 - t) dt$$

→ Antiderivative: $t^3/3 - t^2/2$.

→ $F(3) = [t^3/3 - t^2/2]$ from 0 to 3 = $27/3 - 9/2 = 9 - 4.5 = 4.5$.

→ $F'(x) = x^2 - x$ by the Fundamental Theorem of Calculus (Part 1).

→ $F(3)$ = net signed area under $t^2 - t$ from 0 to 3.

Answer: $F(3) = \frac{9}{2}, \quad F'(x) = x^2 - x$

8. Evaluate using the Fundamental Theorem of Calculus.

$$\int_1^4 \left(\sqrt{x} - \frac{1}{x} \right) dx$$

→ Antiderivative of $\sqrt{x} = x^{1/2}$ is $(2/3)x^{3/2}$.

→ Antiderivative of $1/x$ is $\ln x$.

→ At $x=4$: $(2/3)(8) - \ln 4 = 16/3 - \ln 4$.

→ At $x=1$: $(2/3)(1) - \ln 1 = 2/3$.

→ Result: $(16/3 - \ln 4) - 2/3 = 14/3 - \ln 4$.

→ Note: $14/3 \approx 4.667$ and $\ln 4 \approx 1.386$, so result ≈ 3.28 .

Answer: $\left[\frac{2}{3}x^{3/2} - \ln x \right]_1^4 = \frac{28}{3} - \ln 4$

9. A city's population grows at the rate shown below. The population at $t = 0$ is 50,000. Use the Net Change Theorem to find $P(10)$.

$$P'(t) = 400e^{0.02t}, \quad P(10) = P(0) + \int_0^{10} 400e^{0.02t} dt$$

→ Net Change Theorem: $P(10) = P(0) + \text{integral from 0 to 10 of } P'(t) dt$.

→ Antiderivative of $400e^{0.02t} = 20000e^{0.02t}$.

→ Evaluate: $20000e^{0.2} - 20000 = 20000(e^{0.2} - 1) \approx 4392$.

→ $P(10) \approx 50000 + 4392 = 54,392$ people.

Answer: $P(10) = 50000 + [20000e^{0.02t}]_0^{10} \approx 54392$



10. Compute L_4 , R_4 , and M_4 with $n = 4$ for the integral below. Compare each approximation to the exact value.

$$\int_0^2 x^3 dx, \quad \Delta x = 0.5$$

→ Exact: $[x^4/4]$ from 0 to 2 = $16/4 = 4$.

→ $dx=0.5$. Left pts: 0, 0.5, 1, 1.5 → f : 0, 0.125, 1, 3.375. $L_4=0.5(4.5+...)=2.25$.

→ Right pts: 0.5, 1, 1.5, 2 → f : 0.125, 1, 3.375, 8. $R_4=0.5(12.5)=6.25$.

→ Midpts: 0.25, 0.75, 1.25, 1.75 → f : 0.0156, 0.422, 1.953, 5.359. $M_4 \approx 3.875$.

→ M_4 is closest to the exact value of 4. L_4 underestimates, R_4 overestimates (f is increasing).

Answer: Exact = 4; $L_4 = 2.25$, $R_4 = 6.25$, $M_4 \approx 3.875$

